The Open Source Audit

A Capstone Project for OSS NGMC Course

Units 1 – 5 | Max Marks: 100

Software	MySQL (Database Management System)
License	GNU General Public License v2 (GPL v2)
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Submission	25-03-2026

Introduction

This audit is about MySQL, one of the popular open-source relational databases in the world. You have probably used MySQL without realizing it. It is used by platforms like WordPress, the old YouTube and Twitter. MySQL is also a part of the LAMP stack, which helped make the web more dynamic.

In the following pages I will talk about how MySQL started why it is important what the GPL v2 license means for people who use it every day some questions about open source and big companies, where you can find MySQL on a Linux system the people and tools that work with it and how it compares to its biggest competitor that is not open source.

MySQL has an interesting story. It began as a project in Sweden then became a big part of the web got bought by another company twice and then the person who made it first split it into MariaDB when people started to worry about its future. This story says a lot about what open source can and cannot do.

I am including five shell scripts at the end. Each script shows a skill you can use on Linux explains the ideas behind it and has a place, for a screenshot and the exact command you need to use in WSL.

Part A — Origin and Philosophy (Units 1 & 2)

A1. The Problem MySQL Was Created to Solve

The Web in the Mid-1990s

The web was really taking off in 1994. Websites were starting to need databases to store all sorts of things like user data and product listings. The thing is, the good database systems then were really expensive. Oracle was the one that everyone used. It cost a lot of money to use it. IBM DB2 was just as bad. PostgreSQL was around. It was too complicated for simple websites. There just was not an free option that people could use.

A company from Sweden called TcX was having this problem. The people in charge Michael Widenius also known as Monty and David Axmark were making tools for the web. They kept running into this issue. They needed something that was fast, simple and free, for the people they worked with. They tried to use something called mSQL. It just did not do everything they needed. So in 1995 Monty and David decided to make their database system the database system they were looking for the database system that would solve their problems.

What MySQL Was Built to Do

When MySQL first started it was really important that it worked fast. The people who made MySQL wanted it to be very speedy so they designed it that way from the beginning. MySQL did not have a lot of features but most web applications did not need them at that time. It was good that MySQL could run well on computers. The people who made MySQL decided not to include some things, like stored procedures because they could slow it down. What was most important was that MySQL could keep up with what the users needed. The simple way that MySQL was made worked well because it was what MySQL users wanted.

MySQL got its name from Montys daughter, who was named My and SQL was added to her name. MySQL was first made in 1995. It had a special way of licensing that was different from other software. This special licensing model made MySQL unique, from the start. People could use MySQL for free. Companies had to pay to use it. This way of licensing, where people could use it for free but companies had to pay was used from the beginning. It helped shape how MySQL was developed. After that MySQL developed in two ways.

It Spread Across the Web

MySQL started to get popular in the 1990s and it just kept on growing through the early 2000s. The fact that MySQL was free to use helped it become well known quickly among people who wrote code and wanted something that was not too complicated. Usually it was easy to install MySQL. It worked well for most websites. A lot of people started using MySQL with Linux, Apache and PHP which is often called the LAMP setup. Many companies that hosted websites included MySQL as part of their service. There were many tutorials that explained how to set it up. Big platforms like WordPress, Drupal and Joomla used MySQL from the beginning. The company that made MySQL, MySQL AB said that by 2001, than four million people had installed MySQL.

The Oracle Acquisition and the MariaDB Fork

In the year 2008 Sun Microsystems bought MySQL AB for one billion dollars. When Oracle acquired Sun in the year 2010 they also got MySQL.. That is where things started to get really interesting. Oracle is the company that sells the expensive database that people have to pay for. The people who use open source software were worried: would Oracle let MySQL fade away? Would they change the rules for using MySQL?

Monty Widenius, the person who first created MySQL did not wait to find out what would happen. In the year 2009 before Oracle had even finished buying Sun he made a copy of MySQL. Called it MariaDB. This is what the rules of the GPL version 2 are meant to allow. Because the code for MySQL was open anyone. Including the person who first made it. Could take the code. Keep working on it on their own. Now MariaDB is the default choice of MySQL, in Fedora, Red Hat and Arch Linux. The GPL rules made this possible. No matter what Oracle decides to do with MySQL, the community of users has a copy of the software that they can control.

A2. The Licence — What It Actually Says

MySql Uses Two Types of Licenses

MySQL has a way of doing things with licenses. The Community Edition of MySQL is free for people to use because it is under the GPL v2 license. On the hand the Enterprise Edition of MySQL is not free and people have to pay for it because it is under a commercial license.This means that if you use MySQL to make something and you want to share it with people the GPL v2 license says that you have to make your whole project open source. So companies that want to use MySQL in things they sell to people have to pay Oracle for a license.

MySQL has always done things this way. MySQL is free if you are doing open source work, with it. You have to pay if you are using it for commercial things that you want to keep to yourself.

The Four Freedoms of Free Software

The Free Software Foundation says what free software is by talking about four freedoms. MySQLs GPL v2 gives us all four of these freedoms.

* Freedom 0 is the freedom to use MySQL for anything you want. You can download MySQL. Use it for any purpose.

* Freedom 1 is the freedom to see how MySQL works. You can look at MySQLs source code on GitHub, at github.com/mysql/mysql-server.

* Freedom 2 is the freedom to share MySQL with anyone you want. You can give MySQL to anyone.

* Freedom 3 is the freedom to change MySQL. Then share the new version. This is how MariaDB and Percona Server were made.

Freedom 3 is really important because it saved MySQL from being controlled by one company. Since the GPL says so the community could make a version of MySQL if they did not like the way Oracle was taking care of it. This is what saved MySQL from being locked in by a company. MySQLs Freedom 3 is what kept MySQL free.

Can a Company Take MySQL, Modify It, and Sell It Without Sharing?

If people give MySQL to people that is a different story. The GPL v2 has a rule called copyleft. This rule means that if you share software that is based on MySQLs code you have to share your code too. You cannot change MySQL. Then sell the new version without sharing your code.

There is a way around this rule though. If a company changes MySQL for themselves and does not give it to anyone else they do not have to share those changes. For example Facebook changed MySQL for their use and they did not have to share those changes with anyone. This is okay, under the GPL v2 rules. Facebook was able to use MySQL with their changes and they did not have to make those changes public. This is totally allowed under the GPL v2 rules. MySQL is still protected by these rules and companies have to follow them when they use MySQL.

GPL v2 vs MIT — The Difference

MIT is a flexible licence. It basically says: you can do anything you want with my code just keep the copyright notice. For example a company can take code licensed under MIT change it build a product and never share their changes. Apple did something with BSD Unix when they built macOS.

GPL version 2 is a kind of licence. It helps keep software free. If you share software that has GPL code you must share your version under GPL too. MySQL chose GPL because it wanted companies that use their database to either give back to the community or pay for it. If MySQL

had chosen MIT companies could have taken the code changed it and made products without helping the community.

If I were making a project or library that I wanted many people to use I might choose MIT.. If I were making something that I wanted to stay open and not be controlled by a big company GPL would be the better choice. For a database that many businesses rely on GPL made sense because it also offered a licence option.

Free as in Free Beer vs Free as in Freedom

Stallmans distinction is important: gratis means something is free of charge libre means you have control over the software. MySQL is an example of why we need both.

* MySQL Community Edition is free. You do not have to pay anything for it.

* MySQL Community Edition is also libre. You can look at the source code make changes to it and share it with others under the GPL.

* MySQL Enterprise Edition is not free. You have to pay Oracle for it.. The source code is still available for people to see.

The important thing to remember is that because MySQL is libre the community was able to create a version called MariaDB when Oracle took control of MySQL. If MySQL was only free Oracle could have changed the rules at any time. Everyone would have had to follow them. The freedom in the license gave the MySQL community the power to make their decisions, about the software.

A3. The Ethics of Open Source

Should All Software Be Open Source?

The case for open source software is strong when it comes to software that runs infrastructure. For example a database like MySQL powers healthcare systems, financial platforms and government services. When the code is open anyone can check it for security problems verify that it does what it claims to do and fix issues without having to wait for a vendor. Closed critical infrastructure is a risk that society takes on behalf of one companys profits.

On the hand some people argue that open source does not always work. This is because open source relies on people contributing to it and that does not always happen. For instance the Heartbleed vulnerability in OpenSSL in 2014 affected a lot of websites. Around half a billion. The library was only maintained by two people. Just because the source code is available it does not mean that it is well-maintained or secure. Both things need to be true for source to work well.

In my view the default should be source for infrastructure and widely-used public tools. Commercial software still has a place in innovation.. I think it is hard to justify hiding how critical database infrastructure works, with no way for users to verify or audit it in 2024. Open source software, like MySQL should be used for infrastructure. MySQL is used in areas and open source software can help make it more secure.

Is It Ethical for Companies to Build Profitable Products on Community Labour?

Take MySQL for example. When Oracle bought it for a $7.4 billion things changed slowly. Many volunteers built MySQL over years giving their time and code. This group effort became the base for Oracle's paid product. MySQL Enterprise Edition.. The people who made MySQL's core did not get anything. The money went to Oracle not to those who worked for free.

The rules do allow it. Oracle gives out code under GPL. Still controls MySQL. Many people thought this was wrong so Monty Widenius created MariaDB on his own. This split was not about code differences. It was about a principle: the people who helped build something should control it.

Some companies give back, to source share updates or support creators.. Others take freely and do not give much back. Because of how Oracle handled MySQL people started trusting others. Many Linux versions today use MariaDB instead. It just feels now.

Standing on the Shoulders of Giants

Every web application that uses MySQL is built on a lot of work done by database researchers, SQL standards committees and open source contributors over years.

The person who writes a query like SELECT * FROM users is using relational algebra from the 1970s the SQL standard, the MySQL storage engine, the Linux kernel, the networking stack and the compiler that built all of these things.

Open source software makes it clear who did all this work.

When you can read the source code you can see whose work you are using.

When something is proprietary you do not know who did the work.

MySQL has been a part of the open web for almost thirty years.

MySQL will keep being used whether Oracle or MariaDB is, in charge because the GPL license ensures that it will stay available.

Part B — Linux Footprint (Unit 2)

Installation Method

MySQL is available through APT on Ubuntu and Debian-based systems. The package mysql-server installs the server, mysql-client installs just the command-line tools

```
aditya@aditya:~$       mysql --version
mysql  Ver 8.0.45-0ubuntu0.24.04.1 for Linux on x86_64 ((Ubuntu))
aditya@aditya:          ~$
```

On Fedora or RHEL systems we use dnf as the package manager.

MySQL has its official repositories for people who want the latest MySQL version, not the one that comes with the operating system.

You can also get the MySQL source code, from github.com/mysql/mysql-server if you want to build it yourself.

Key Directories on a Linux System

When MySQL is installed, its files are spread across standard directories following the Filesystem Hierarchy Standard:

Path	What it contains
/etc/mysql/	Configuration files. The main config is /etc/mysql/mysql.conf.d/mysqld.cnf
/var/lib/mysql/	The actual database data files. Each database is a subdirectory here.
/var/log/mysql/	Log files including the error log at /var/log/mysql/error.log
/usr/bin/mysql	The MySQL client binary you use on the command line
/usr/sbin/mysqld	The MySQL server daemon — the actual database server process
/usr/share/mysql/	Support files, default config templates, timezone data
/run/mysqld/	Runtime files including the Unix socket file mysqld.sock

```
aditya@aditya:~$          sudo ls -l /var/lib/mysql/
total 91604
-rw-r----- 1 mysql mysql   196608 Mar 24 14:22 '#ib_16384_0.dblwr'
-rw-r----- 1 mysql mysql  8585216 Mar 24 14:19 '#ib_16384_1.dblwr'
drwxr-x--- 2 mysql mysql     4096 Mar 24 14:20 '#innodb_redo'
drwxr-x--- 2 mysql mysql     4096 Mar 24 14:20 '#innodb_temp'
-rw-r----- 1 mysql mysql        5 Mar 24 14:20  aditya.pid
-rw-r----- 1 mysql mysql       56 Mar 24 14:19  auto.cnf
-rw-r----- 1 mysql mysql      180 Mar 24 14:20  binlog.000001
-rw-r----- 1 mysql mysql      404 Mar 24 14:20  binlog.000002
-rw-r----- 1 mysql mysql      157 Mar 24 14:20  binlog.000003
-rw-r----- 1 mysql mysql       48 Mar 24 14:20  binlog.index
-rw------- 1 mysql mysql     1705 Mar 24 14:20  ca-key.pem
-rw-r--r-- 1 mysql mysql     1112 Mar 24 14:20  ca.pem
-rw-r--r-- 1 mysql mysql     1112 Mar 24 14:20  client-cert.pem
-rw------- 1 mysql mysql     1705 Mar 24 14:20  client-key.pem
-rw-r--r-- 1 root  root         0 Mar 24 14:20  debian-5.7.flag
-rw-r----- 1 mysql mysql     3438 Mar 24 14:20  ib_buffer_pool
-rw-r----- 1 mysql mysql 12582912 Mar 24 14:20  ibdata1
-rw-r----- 1 mysql mysql 12582912 Mar 24 14:20  ibtmp1
drwxr-x--- 2 mysql mysql     4096 Mar 24 14:20  mysql
-rw-r----- 1 mysql mysql 26214400 Mar 24 14:20  mysql.ibd
drwxr-x--- 2 mysql mysql     4096 Mar 24 14:20  performance_schema
-rw------- 1 mysql mysql     1705 Mar 24 14:20  private_key.pem
-rw-r--r-- 1 mysql mysql      452 Mar 24 14:20  public_key.pem
-rw-r--r-- 1 mysql mysql     1112 Mar 24 14:20  server-cert.pem
-rw------- 1 mysql mysql     1705 Mar 24 14:20  server-key.pem
drwxr-x--- 2 mysql mysql     4096 Mar 24 14:20  sys
-rw-r----- 1 mysql mysql 16777216 Mar 24 14:22  undo_001
-rw-r----- 1 mysql mysql 16777216 Mar 24 14:22  undo_002
aditya@nethaniet:~$
```

User and Group — Why It Matters for Security

MySQL runs as an user called 'mysql' not as the main administrator.

This is a rule in Linux for keeping things secure: a service should only be able to do what it needs to do.

The mysql user is in charge of the folder /var/lib/mysql and a special connection point in /run/mysqld/.

It can't mess with system files like /etc/shadow can't change system programs and can't get more power than it already has.

This is important because even if someone finds a weakness, in the MySQL server they only get the mysql users powers. Not the administrators.

They can look at the database files. Can't take over the whole system unless they do another attack.

The fact that MySQL is source means that security experts can read the code and find weaknesses before bad people do.

The mysql user and MySQL server are designed to work securely.

MySQLs security and the way it runs help protect the system.

```
aditya@aditya:~$          id mysql
uid=118(mysql) gid=125(mysql) groups=125(mysql)
aditya@aditya          :~$ sudo ls -la /var/lib/mysql/ | head -8
total 91612
-rw-r-----  1 mysql mysql   196608 Mar 24 14:22 #ib_16384_0.dblwr
-rw-r-----  1 mysql mysql  8585216 Mar 24 14:19 #ib_16384_1.dblwr
drwxr-x---  2 mysql mysql     4096 Mar 24 14:20 #innodb_redo
drwxr-x---  2 mysql mysql     4096 Mar 24 14:20 #innodb_temp
drwx------  7 mysql mysql     4096 Mar 24 14:20 .
drwxr-xr-x 67 root  root      4096 Mar 24 14:19
-rw-r-----  1 mysql mysql        5 Mar 24 14:20 aditya.pid
aditya@aditya:          ~$ ps aux | grep mysqld
mysql       3010  1.7 10.1 2376312 392376 ?      Ssl  14:20   0:08 /usr/sbin/mysqld
nethani+    3381  0.0  0.0   4092  1920 pts/0    S+   14:28   0:00 grep --color=auto mysqld
aditya@aditya:          ~$
```

Service Management

Running MySQL usually means it works behind the scenes like a background helper. To contro l it- whether waking it up, pausing, resetting, or seeing if it's active - you rely on familiar syste mctl tools found in most Linux systems.

```
aditya@aditya:~$          sudo systemctl status mysql && sudo systemctl is-enabled mysql
● mysql.service - MySQL Community Server
     Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-03-24 14:20:50 +03; 9min ago
    Process: 3002 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
   Main PID: 3010 (mysqld)
     Status: "Server is operational"
      Tasks: 37 (limit: 4504)
     Memory: 349.6M (peak: 378.9M)
        CPU: 9.194s
     CGroup: /system.slice/mysql.service
             └─3010 /usr/sbin/mysqld

Mar 24 14:20:48 aditya systemd[1]: Starting mysql.service - MySQL Community Server...
Mar 24 14:20:50 aditya systemd[1]: Started mysql.service - MySQL Community Server.
enabled
aditya@aditya:~$
```

Right after installation, it kicks in when the system boots up. When tweaks happen in /etc/mysql/,
a reboot of the process is needed. Watching errors as they appear? Try running sudo tail -f /var/log/mysql/error.log.

Update Model

Fresh security updates are coming through the repository.

I run `sudo update` and then `sudo apt upgrade` to get the latest fixes from the distribution teams.

MySQL code is open so people on the internet can find problems talk about them and see fixes being reviewed by anyone who wants to check.

This is different, from database tools where none of that openness exists.

Part C — The FOSS Ecosystem (Units 3 & 4)

What MySQL Depends On

• OpenSSL. Helps to keep client-server connections with encryption. When you connect to a MySQL server away using SSL/TLS OpenSSL handles the encryption.

- zlib. Helps to make data smaller during transfer, over the network and when stored.
- GNU C Library (libc). A part of the C programming language that MySQL uses on Linux.
- CMake. A tool used to build MySQL from its code.
- Boost C++ Libraries. Used in MySQLs code for tools and ways to organize data.
- Linux kernel. MySQL uses the kernel to work with files manage memory and connect to the network.

```
aditya@aditya:~$                ldd $(which mysql) | head -15
        linux-vdso.so.1 (0x00007e907f177000)
        libedit.so.2 => /lib/x86_64-linux-gnu/libedit.so.2 (0x00007e907f127000)
        libz.so.1 => /lib/x86_64-linux-gnu/libz.so.1 (0x00007e907f10b000)
        libzstd.so.1 => /lib/x86_64-linux-gnu/libzstd.so.1 (0x00007e907e946000)
        libssl.so.3 => /lib/x86_64-linux-gnu/libssl.so.3 (0x00007e907e89c000)
        libcrypto.so.3 => /lib/x86_64-linux-gnu/libcrypto.so.3 (0x00007e907e200000)
        libresolv.so.2 => /lib/x86_64-linux-gnu/libresolv.so.2 (0x00007e907f0f6000)
        libstdc++.so.6 => /lib/x86_64-linux-gnu/libstdc++.so.6 (0x00007e907de00000)
        libgcc_s.so.1 => /lib/x86_64-linux-gnu/libgcc_s.so.1 (0x00007e907f0c8000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007e907da00000)
        libtinfo.so.6 => /lib/x86_64-linux-gnu/libtinfo.so.6 (0x00007e907e868000)
        libbsd.so.0 => /lib/x86_64-linux-gnu/libbsd.so.0 (0x00007e907f0b2000)
        libm.so.6 => /lib/x86_64-linux-gnu/libm.so.6 (0x00007e907e77f000)
        /lib64/ld-linux-x86-64.so.2 (0x00007e907f179000)
        libmd.so.0 => /lib/x86_64-linux-gnu/libmd.so.0 (0x00007e907f0a1000)
aditya@aditya:~$
```

What MySQL Has Enabled

The LAMP Stack

MySQL is the part of something called LAMP. LAMP stands for Linux, Apache, MySQL and PHP or Python or Perl. This group of things became the way to make websites from the year

2000 to around the middle of the 2010s. People still use it a lot today. WordPress is a deal because it runs around 43 percent of all the websites on the internet right now. WordPress uses MySQL or something similar called MariaDB to store all its information. Other things, like Drupal and Joomla were made in a way. They all use MySQL to get things done.

Forks and Derivatives

MariaDB is the most significant fork, created by MySQL's original developer in 2009. It is now the default in Fedora, Red Hat Enterprise Linux and Arch Linux. MariaDB maintains wire-compatibility with MySQL — applications written for MySQL work with MariaDB unchanged — while adding its own storage engines and features.

Percona Server is a performance-focused MySQL fork used in high-traffic production environments. Percona also maintains XtraBackup, a widely used open source hot-backup tool for MySQL. TiDB is a modern distributed SQL database that is MySQL-compatible at the protocol level, meaning existing MySQL applications can point to it without code changes.

Tools Built Around MySQL

phpMyAdmin is a tool that helps us manage MySQL databases on the web. It is source and people have been using it for a long time to manage databases using a browser. MySQL Workbench is the tool that we use to manage databases with a graphical user interface.

There are also frameworks like SQLAlchemy for Python and Hibernate for Java. These frameworks are called ORM frameworks. They let developers work with MySQL databases using object-oriented code. We can use MySQL with lots of web frameworks, like Django, Laravel, Rails, Spring and Express. Each of these frameworks has a connector that lets us use MySQL with them.

MySQL in the LAMP Stack

MySQL is really important in LAMP because it stores and retrieves data that is organized in a way. So here is what happens when you make a request. Your browser sends a request. Apache gets this request. Then it passes it to PHP. Then PHP asks MySQL for some data, like a blog post and the comments people made on it. MySQL gives PHP the data it asked for. PHP takes this data. Formats it into HTML. Then Apache sends this HTML back to your browser. MySQL is right, in the middle of all this. That is why MySQL query performance is usually the thing that slows down web applications. MySQL query performance is the thing to focus on.

Community and Governance

MySQL is a bit of a mess because Oracle owns it. Oracle is in charge of the MySQL name. Makes sure it gets updated. If you want to report a problem with MySQL you can do that at bugs.mysql.com. These days the main place for people who care about MySQL to get together

is the Percona Live conference because Oracle does not have a conference just for MySQL anymore.

There is a group called the MariaDB Foundation that takes care of MariaDB. They make a plan for what they want to do with MariaDB. They ask the community for ideas on what features to add. The reason MariaDB is separate from MySQL is that people did not trust Oracle when they bought MySQL. A lot of Linux distributions are choosing MariaDB as the default database. This is like a vote of confidence from the community. They are choosing MariaDB over MySQL, which says a lot, about what the community thinks about MySQL under Oracle and MariaDB under the foundation.

Part D — Open Source vs Proprietary (Critical Analysis)

Comparison: MySQL vs Oracle Database

Dimension	MySQL (Open Source)	Oracle DB (Proprietary)
Cost	Free. MySQL Community Edition costs nothing to download and use.	Oracle Enterprise Edition costs tens of thousands of dollars per processor annually.
Security (Code Audit)	Source is public on GitHub. Anyone can find and report bugs. Community patches are fast.	Closed source. Security depends entirely on Oracle's internal team.
Support & Reliability	Community forums, Stack Overflow, official docs. No SLA on the free edition.	Paid support with guaranteed SLA. Enterprise-grade accountability.
Freedom to Modify	GPL v2 lets you read and modify the source. MariaDB and Percona exist because of this.	No source access. You use what Oracle ships. No forking allowed.
Community vs Corporate Control	Technically owned by Oracle, but MariaDB exists as a community-controlled alternative under the GPL.	100% Oracle controlled. Pricing, features, and roadmap decided by one company.
Overall Verdict	Right choice for nearly all web applications. Free, battle-tested, huge community.	Worth it only in very large enterprises that need Oracle support contracts.

Verdict

For websites and companies MySQL is a good choice. It does not cost anything it is easy to learn about. A lot of people use it. A lot of people have used MySQL. It works well. Some people are worried about the company Oracle being in charge of MySQL. Mysql is protected by a special kind of license. This license is called the GPL license. Because of this license people can make their versions of MySQL like MariaDB. MariaDB is similar to MySQL so people can use it instead if they want to.

Would I help with MySQL or MariaDB? Yes I would. MariaDB is a kind of project that people work on together and it has a group that supports it. If I helped with MariaDB even if it was a little bit it would be helpful. I could tell them about a problem help them write instructions or help them test things. This kind of work is important because a lot of websites use MySQL or MariaDB. The good thing about open source projects, like MariaDB is that anyone can help.

Part E — Shell Script Tasks (Units 2 & 5)

All five scripts were written and tested on Ubuntu 22.04 LTS under WSL2 on Windows 11. Each script is shown below with its concepts, full code, and a screenshot placeholder with the exact command to run.

Script 1 — System Identity Report

Concepts Used

- Variables — STUDENT_NAME and SOFTWARE are plain variables. KERNEL, DISTRO,LOGGED_IN, UPTIME, DATE all use $() to capture live system output.
- Command substitution $() — uname -r gets kernel version, lsb_release -d gets the distro, whoamigets current user, uptime -p gives readable uptime.
- echo — all output goes through echo with simple divider lines.
- cut -f2 — strips the label from lsb_release output and keeps just the distro name.

```bash
#!/bin/bash
# Script 1 - System Identity Report
# Name: [ ADITYA JAYANT] | Reg: [24BAI10579]
# Course: Open Source Software

# storing my details in variables
STUDENT_NAME="[ADITYA JAYANT]"
SOFTWARE="MySQL"

# command substitution to grab live system info
KERNEL=$(uname -r)
DISTRO=$(lsb_release -d | cut -f2)
LOGGED_IN=$(whoami)
UPTIME=$(uptime -p)
DATE=$(date)

echo "==================================="
echo "  System Info - OSS Audit"
echo "==================================="
echo "Student   : $STUDENT_NAME"
echo "Software  : $SOFTWARE"
echo ""
echo "Distro    : $DISTRO"
echo "Kernel    : $KERNEL"
echo "User      : $LOGGED_IN"
echo "Home      : $HOME"
echo "Uptime    : $UPTIME"
echo "Date/Time : $DATE"
echo ""
echo "-----------------------------------"
echo "This system runs Linux, licensed"
echo "under GPL v2. Free to use, study,"
echo "modify and share."
echo "==================================="
```

```
aditya@aditya:~/vityarthi-apache-mysql$          sudo chmod +x script1_system_identity.sh && ./script1_system_identity.sh
═══════════════════════════════════════
  System Info - OSS Audit
═══════════════════════════════════════
Software  : MySQL
Distro    : Ubuntu 24.04.4 LTS
Kernel    : 6.6.87.2-microsoft-standard-WSL2
User      : aditya
Home      : /home/aditya
Uptime    : up 26 minutes
Date/Time : Tue Mar 24 14:39:46 +03 2026

─────────────────────────────────────
This system runs Linux, licensed
under GPL v2. Free to use, study,
modify and share.
═══════════════════════════════════════
aditya@aditya  .:~/Vityarthi-Apache-Mysql$
```

This script prints a live snapshot of the system. Every value is read with command substitution so the output is always accurate. The GPL note at the end ties it back to the course — the OS itself is open source under the same licence we studied.

Script 2 — FOSS Package Inspector

Concepts Used

- if-else — checks if mysql-server is installed with dpkg -l. Shows version if yes, install instructions ifno.
- case statement — maps the package name to a short note about its open source history. Coversmysql, mariadb, apache2 and a default fallback.
- dpkg -l — queries the Debian/Ubuntu package database to check if a package is installed.
- mysql --version — called inside the if block to display the installed version.

```
#!/bin/bash
# Script 2 - FOSS Package Inspector
# Name: ADITYA JAYANT | Reg: 24BAI10579
# Course: Open Source Software

PACKAGE="mysql-server"

echo "Checking if $PACKAGE is installed..."
echo ""

# if-else to see if the package is there
if dpkg -l "$PACKAGE" &>/dev/null; then
    echo "$PACKAGE is installed"
    echo ""
    # show version info
    mysql --version
    echo "License : GPL v2"
else
    echo "$PACKAGE is not installed"
    echo "Install it with: sudo apt install mysql-server"
fi

echo ""
echo "--- Philosophy note ---"

# case statement - note based on package name
case $PACKAGE in
    mysql-server|mysql)
        echo "MySQL started as a lightweight database in 1995."
        echo "It became one of the most used databases in the world."
        echo "Sun bought it, then Oracle did - and the community forked to MariaDB."
        ;;
    mariadb)
        echo "MariaDB is what happens when a community refuses to lose control."
        ;;
    apache2|httpd)
        echo "Apache proved open source could run the commercial web."
```

```
        ;;
    apache2|httpd)
        echo "Apache proved open source could run the commercial web."
        ;;
    *)
        echo "Open source - built by people who believed in sharing."
        ;;
esac
```

```
aditya@aditya:~/vityarthi-apache-mysql$          chmod +x script2_package_inspector.sh && ./script2_package_inspector.sh
Checking if mysql-server is installed...

mysql-server is installed

mysql  Ver 8.0.45-0ubuntu0.24.04.1 for Linux on x86_64 ((Ubuntu))
License : GPL v2

--- Philosophy note ---
MySQL started as a lightweight database in 1995.
It became one of the most used databases in the world.
Sun bought it, then Oracle did - and the community forked to MariaDB.
aditya@aditya
```

Simple defensive scripting — check first, then act. The case statement covers the main software options from the assignment brief. The MySQL note specifically mentions the Oracle and MariaDB story because that is the most important thing about MySQL's open source history.

Script 3 — Disk and Permission Auditor

Concepts Used

- for loop — loops over the DIRS array and runs the same checks on each directory.
- Bash array — DIRS=(...) holds the list of paths, ${DIRS[@]} expands all of them in the loop.
- [-d] test — checks the directory exists before trying ls or du on it.
- ls -ld with awk — ls -ld gives the full metadata line, awk pulls out permissions and owner fields.
- du -sh with cut — du -sh gives human-readable size, cut -f1 keeps just the number.
- Second loop — reuses the for pattern to check MySQL-specific paths at the end.

```bash
#!/bin/bash
# Script 3 - Disk and Permission Auditor
# Name: ADITYA JAYANT | Reg: 24BAI10579
# Course: Open Source Software

# directories i want to check
DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit"
echo "---------------"

# loop through each one
for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        # awk pulls permissions and owner out of ls output
        PERMS=$(ls -ld "$DIR" | awk '{print $1, $3, $4}')
        SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)
        echo "$DIR"
        echo "  Permissions : $PERMS"
        echo "  Size        : $SIZE"
        echo ""
    else
        echo "$DIR does not exist"
        echo ""
    fi
done

# checking mysql specific paths
echo "--- MySQL config paths ---"
```

```bash
# checking mysql specific paths
echo "--- MySQL config paths ---"

MYSQL_PATHS=("/etc/mysql" "/var/lib/mysql" "/var/log/mysql")

for P in "${MYSQL_PATHS[@]}"; do
    if [ -e "$P" ]; then
        echo "Found: $P"
        ls -ld "$P"
    else
        echo "Not found: $P"
    fi
done
```

```
aditya@aditya:~/vityarthi-apache-mysql$            chmod +x script3_disk_auditor.sh && ./script3_disk_auditor.sh
Directory Audit
────────────────

/etc
  Permissions : drwxr-xr-x root root
  Size        : 7.5M

/var/log
  Permissions : drwxrwxr-x root syslog
  Size        : 38G

/home
  Permissions : drwxr-xr-x root root
  Size        : 17G

/usr/bin
  Permissions : drwxr-xr-x root root
  Size        : 1.3G

/tmp
  Permissions : drwxrwxrwt root root
  Size        : 76K

--- MySQL config paths ---
Found: /etc/mysql
drwxr-xr-x 4 root root 4096 Mar 24 14:19 /etc/mysql
Found: /var/lib/mysql
drwx────── 7 mysql mysql 4096 Mar 24 14:20 /var/lib/mysql
Found: /var/log/mysql
drwxr-x--- 2 mysql adm 4096 Mar 24 14:19 /var/log/mysql
aditya@aditya
```

The for loop is the core idea — same logic applied to multiple directories without repeating code. The

MySQL path check at the end is a second loop doing a slightly different job, showing that the same construct works for different situations in one script.

Script 4 — Log File Analyzer

Concepts Used

- $1 and $2 arguments — log file and keyword passed on the command line. ${2:-'error'} sets adefault if nothing is given.
- [! -f] file test — exits with a clear message if the file does not exist.
- while IFS= read -r — reads the file line by line without breaking on spaces or backslashes.
- if inside while — each line is tested with grep -iq (case insensitive). COUNT goes up by one eachmatch.
- Counter arithmetic — COUNT=$((COUNT + 1)) increments the counter in bash.
- Retry loop — if the file is empty, waits 2 seconds and tries again up to 3 times.
- grep + tail -5 — finds all matching lines then keeps only the last 5.

```
#!/bin/bash
# Script 4 - Log File Analyzer
# Name: ADITYA JAYANT | Reg: 24BAI10579
# Course: Open Source Software
# Usage: ./script4_log_analyzer.sh /var/log/syslog error

# log file and keyword from arguments
LOGFILE=$1
KEYWORD=${2:-"error"}   # defaults to 'error' if nothing passed
COUNT=0
RETRY=0

echo "Log Analyzer"
echo "File    : $LOGFILE"
echo "Keyword : $KEYWORD"
echo ""

# check the file exists first
if [ ! -f "$LOGFILE" ]; then
    echo "Error: $LOGFILE not found"
    echo "Try: sudo ./script4_log_analyzer.sh /var/log/syslog error"
    exit 1
fi

# wait and retry if file is empty
while [ ! -s "$LOGFILE" ] && [ $RETRY -lt 3 ]; do
    echo "File is empty, waiting 2 seconds..."
    sleep 2
    RETRY=$((RETRY + 1))
done
```

```
# wait and retry if file is empty
while [ ! -s "$LOGFILE" ] && [ $RETRY -lt 3 ]; do
    echo "File is empty, waiting 2 seconds..."
    sleep 2
    RETRY=$((RETRY + 1))
done

# read the file line by line
while IFS= read -r LINE; do
    # check each line for the keyword
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

echo "Found '$KEYWORD' $COUNT times in $LOGFILE"
echo ""
echo "Last 5 matching lines:"
echo "----------------------"
grep -i "$KEYWORD" "$LOGFILE" | tail -5
```

```
aditya@aditya:~/vityarthi-apache-mysql$          chmod +x script4_log_analyzer.sh && ./script4_log_analyzer.sh /var/log/dpkg.log error
Log Analyzer
File    : /var/log/dpkg.log
Keyword : error

Found 'error' 0 times in /var/log/dpkg.log

Last 5 matching lines:
----------------------
aditya@aditya:          ~/Vityarthi-Apache-Mysql$
```

The while-read loop is the correct way to read a file line by line in bash. Making the log path and keyword into arguments means the same script works for the MySQL error log at /var/log/mysql/error.log as well as system logs — useful for the actual software being audited.

Script 5 — Open Source Manifesto Generator

Concepts Used

- read -p — prompts the user and stores the answer. Used three times for TOOL, FREEDOM andBUILD.
- Variables in echo — $TOOL, $FREEDOM and $BUILD drop into the manifesto text to personaliseit.
- Grouped output with {} and > — all echo lines are wrapped in {} and sent to the file with one >.
- date command — TODAY=$(date '+%d %B %Y') gives a readable date for the header.
- $(whoami) in filename — creates a filename unique to the current user.

• cat — displays the saved file immediately after writing.Alias concept — shown in the comment at the top with a real example of how bash aliases work.

```bash
#!/bin/bash
# Script 5 - Open Source Manifesto Generator
# Name: ADITYA JAYANT | Reg: 24BAI10579
# Course: Open Source Software

# note on aliases: you can do alias today='date +%d\ %B\ %Y' in bash
# i'm doing the same idea here with a variable instead
TODAY=$(date '+%d %B %Y')

echo "Open Source Manifesto Generator"
echo "--------------------------------"
echo "Answer 3 questions:"
echo ""

# reading input from the user
read -p "1. One open source tool you use daily: " TOOL
read -p "2. What does freedom mean to you (one word): " FREEDOM
read -p "3. What would you build and share freely: " BUILD

# output file named after the current user
OUTPUT="manifesto_$(whoami).txt"

# writing to file - all echo lines grouped with {}
{
echo "My Open Source Manifesto"
echo "Written by: $(whoami) on $TODAY"
echo ""
echo "Every day I use $TOOL, built by someone who decided"
echo "to share their work instead of locking it away."
echo ""
echo "To me, freedom means $FREEDOM. That is what the GPL is"
echo "trying to protect. MySQL is licensed under GPL v2,"
echo "which is why even after Oracle acquired it, the"
echo "community could fork it into MariaDB."
echo ""
echo "If I built something, I would create $BUILD"
```

```
echo "If I built something, I would create $BUILD"
echo "and release it openly. Because the tools I rely on"
echo "were shared freely, and I think that matters."
echo ""
echo "Open source is not just a licence. It is a mindset."
} > "$OUTPUT"

echo ""
echo "Saved to $OUTPUT"
echo ""
cat "$OUTPUT"
```

•

```
aditya@aditya:~/vityarthi-apache-mysql$          chmod +x script5_manifesto_generator.sh && ./script5_manifesto_generator.s
h
Open Source Manifesto Generator
──────────────────────────────

Answer 3 questions:

1. One open source tool you use daily: MySQL
2. What does freedom mean to you (one word): Choice
3. What would you build and share freely: Knowledge

Saved to manifesto_adityal.txt

My Open Source Manifesto
Written by: aditya on 24 March 2026

Every day I use MySQL, built by someone who decided
to share their work instead of locking it away.

To me, freedom means Choice. That is what the GPL is
trying to protect. MySQL is licensed under GPL v2,
which is why even after Oracle acquired it, the
community could fork it into MariaDB.

If I built something, I would create Knowledge
and release it openly. Because the tools I rely on
were shared freely, and I think that matters.

Open source is not just a licence. It is a mindset.
aditya@aditya          .:~/Vityarthi-Apache-Mysql$
```

The grouped {} redirect is cleaner than appending line by line with >>. The manifesto text mentions MySQL and MariaDB directly in the middle paragraph, which ties the creative output back to the software being studied rather than being generic.

Conclusion

MySQL has one of those stories you just can't ignore.

It started as a solution for a big problem. Back then nobody had a free database that worked well for building websites.

Before long MySQL became the foundation for the internet.

Then things got interesting: Oracle, a company that makes money from databases bought it.

The open source community didn't like that so they created a version of MySQL.

Thanks to the GPL v2 license they were able to do that.

That whole series of events. Create something share it watch it become popular get bought and survive because of the license. Feels like a guide for how open source projects deal with the business world.

The license isn't paperwork; it's what gives the community real power when things get tough.

Here are five scripts in this project.

They show off Linux skills using tools like

* bash* grep* awk* du* dpkg.

Every one of these tools is source.Somebody decided to share their work. Thats why we can use them today.MySQL is one part of this huge network of shared software. It's what makes modern computing work.If you work in software understanding how this network works and what keeps it together (those licenses) is really important.MySQL and open source licenses are key, to making all this work.The open source community and MySQL are closely tied.MySQLs story is an example of how open source projects can succeed.

References

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